

MARCH AFB, CA, USA

WMO: 722860

Lat: 33.900N

Lon: 117.250W

Elev: 1536

StdP: 13.90

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.5	34.4	1.3	6.2	60.9	6.6	8.1	61.6	24.8	54.8	20.7	54.6	1.4	320	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	31.6	101.4	67.6	98.8	66.7	95.8	66.1	71.9	93.1	70.5	91.3	69.2	89.1	10.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	102.9	75.4	64.2	95.2	74.2	63.0	91.1	73.0	36.5	92.8	35.2	90.9	34.1	89.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.9	15.7	13.2	DB	25.4	107.2	4.5	2.7	22.2	109.2	19.5	110.8	17.0	112.3	13.7	114.3
			WB	22.8	74.6	4.0	1.9	20.0	76.0	17.7	77.1	15.4	78.2	12.5	79.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	52.8	52.8	56.7	59.7	65.1	71.6	77.7	78.9	75.0	66.2	57.8	51.7	(6)
	DBStd	11.17	5.68	5.81	5.61	6.17	6.10	5.52	4.68	5.11	6.19	6.01	6.01	5.45	(7)
	HDD50	138	36	33	9	4	0	0	0	0	0	0	10	46	(8)
	HDD65	1944	378	344	262	179	74	10	0	4	57	224	412		(9)
	CDD50	5213	123	110	218	296	468	647	858	897	749	502	245	100	(10)
	CDD65	1543	1	1	5	21	76	207	393	432	303	94	10	0	(11)
	CDH74	21579	94	119	325	696	1311	2849	4895	5438	3760	1602	440	50	(12)
	CDH80	11479	10	18	80	252	586	1478	2804	3253	2120	740	136	2	(13)
(14) Wind	WSAvg	5.3	4.7	5.0	5.0	5.8	6.1	6.0	6.0	5.6	5.1	4.9	4.5	4.8	(14)
(15) Precipitation	PrecAvg	11.6	2.4	3.0	1.6	0.7	0.2	0.1	0.2	0.1	0.2	0.4	1.0	1.7	(15)
	PrecMax	22.2	11.3	11.2	6.6	2.4	2.0	0.7	1.5	0.4	1.5	5.5	2.1	8.5	(16)
	PrecMin	4.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	(17)
	PrecStd	5.3	2.9	2.5	1.5	0.6	0.4	0.2	0.3	0.1	0.3	1.0	0.5	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.2	82.8	87.3	92.6	97.4	102.4	104.0	105.5	103.7	97.6	89.2	78.9	(19)
		MCWB	54.3	54.9	58.6	58.9	64.0	65.9	69.7	69.0	66.7	61.8	57.3	52.5	(20)
	2%	DB	76.7	77.9	81.6	86.5	91.4	96.6	99.4	101.6	99.3	92.5	84.1	74.8	(21)
		MCWB	52.2	53.5	57.3	58.0	62.3	65.4	68.0	67.7	66.4	59.7	55.7	51.0	(22)
	5%	DB	72.4	72.5	77.3	82.1	86.9	92.7	96.6	98.7	95.9	88.5	79.3	70.3	(23)
		MCWB	51.1	52.2	55.9	57.0	61.0	64.9	67.7	67.2	65.8	59.1	53.9	49.7	(24)
	10%	DB	67.9	67.0	72.7	77.2	81.6	89.1	93.3	95.3	91.9	83.7	74.1	65.5	(25)
		MCWB	49.7	50.7	54.5	55.9	60.6	64.4	67.2	66.7	65.1	58.8	52.3	49.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	59.9	63.4	64.1	67.4	70.9	73.2	74.4	73.0	68.8	61.1	59.6	(27)
		MCDB	66.4	67.4	78.0	82.2	90.9	93.1	94.0	92.0	92.9	79.8	68.8	63.8	(28)
	2%	WB	56.8	57.1	60.5	61.5	65.2	68.4	71.4	72.0	71.0	65.9	58.9	56.7	(29)
		MCDB	64.3	66.5	73.7	78.5	85.1	89.9	92.8	94.0	90.5	80.2	71.8	62.8	(30)
	5%	WB	54.8	55.3	58.6	59.4	63.6	67.0	70.2	70.3	69.2	63.6	57.3	54.4	(31)
		MCDB	62.6	65.5	70.9	74.6	81.8	87.3	91.0	92.3	86.7	77.2	70.3	60.7	(32)
	10%	WB	53.1	53.7	56.6	57.5	61.8	65.6	68.9	68.8	67.6	61.7	55.7	52.2	(33)
		MCDB	61.6	62.4	67.9	71.1	77.5	84.2	88.8	89.9	85.3	74.4	68.0	59.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	25.8	24.6	27.0	28.1	27.1	30.2	30.9	31.6	30.3	29.5	29.0	25.7	(35)
		MCDBR	35.5	37.0	37.1	38.9	37.7	37.2	34.1	34.6	35.4	39.3	38.8	35.1	(36)
	5% WB	MCWBR	16.9	17.2	16.7	15.7	14.0	12.9	11.7	12.3	12.3	14.9	17.1	16.8	(37)
		MCWBR	24.4	25.9	30.3	31.5	32.2	32.8	30.4	30.7	29.4	29.0	29.5	22.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.290	0.299	0.319	0.333	0.346	0.347	0.375	0.359	0.341	0.327	0.297	0.298	(40)
		taud	2.573	2.558	2.482	2.449	2.444	2.454	2.437	2.467	2.503	2.515	2.588	2.578	(41)
	Ebn at Noon		296	304	304	302	298	296	287	291	293	289	291	286	(42)
		Edh at Noon	25	28	33	36	36	36	36	35	32	29	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1033	1318	1774	2158	2370	2545	2426	2291	1960	1501	1148	935	(44)
	RadStd		70	126	86	92	121	124	114	65	63	88	55	64	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	-1.89	-3.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		+1.13	N/A	-3.04	+1.26	N/A	N/A	N/A	-196	+400	+205	

Nomenclature: See separate page